

Q1: Let's consider the operation of a learning switch in the context of a network in which 6 nodes labeled A through F are connected into an Ethernet switch through star topology. Consider MAC and IP addresses of the nodes denoted by MAC-(node label) and IP-(node label). For example, node A in the network will have mac address and ip address denoted as MAC-A and IP-A respectively.

Suppose the following events occur in the network

1. B sends a frame to E
2. E replies with a frame to B
3. A sends a frame to B
4. B replies with a frame to A.

The switch table and the ARP tables of all the nodes are initially empty.

1. Show the state of the switch table after the Event1 and Event4 above.

Switch Table After Event1

Switch Table After Event4

2. Show the ARP tables of each of the nodes of the network.

ARP table for node A

ARP table for node B

ARP table for node C

ARP table for node D

ARP table for node E

ARP table for node F

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Computer Networks	Course Code:	CS307
Program:	BS(Computer Science)	Semester:	Fall 2019
Duration:	20 Minutes	Total Marks:	15
Date:	4 December, 2019	Quiz:	5
Section:	E	Page(s):	1

Name _____

Roll No. _____

- 1) At which layer of the network layer stack do each of the following protocols operate? (Give BOTH layer name and number) [10]

Protocol	Layer
HTTP	Application Layer (5/7)
DHCP	Application Layer (5/7)
Ethernet	Data link layer (2)
DNS	Application Layer (5/7)
IP	Network Layer (3)
ICMP	Network Layer (3)
UDP	Transport layer (4)
IPv6	Network Layer (3)
ARP	Data link layer (2)
FTP	Application Layer (5/7)

- 2) What is Encapsulation? Describe the process of encapsulation when a packet moves from Transport layer to Network Layer on a sending host. [5]

The term encapsulation refers to a process in which protocol information is added to the data when it passes through the layers.

Transport layer breaks the received data stream from upper layers into smaller pieces. Next, it creates a header for each data piece. This header contains all necessary

information about the piece that the transport layer in remote host needs to reassemble the data stream back from the pieces. Once header is attached, data piece is referred as segment. Once segments are created, they are handed down to the network layer for further processing. Network layer creates a header for each received segment from transport layer.

National University of Computer and Emerging Sciences, Lahore Campus
Quiz5 [BS(CS): Section D] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: December 5, 2022

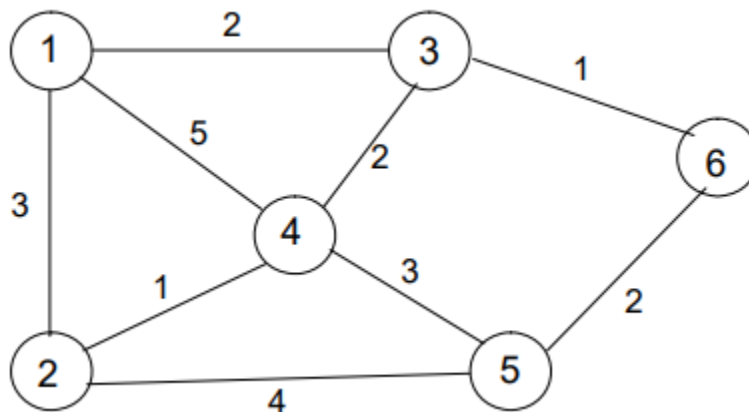
Total Marks: 10

Duration: 20 -Minutes

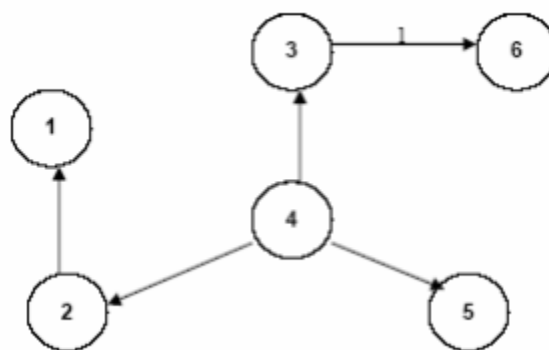
Name ----- Roll #----- Section -----

Instructions: Answer the question(s) on this sheet. You can make use of rough sheet (not to be attached).

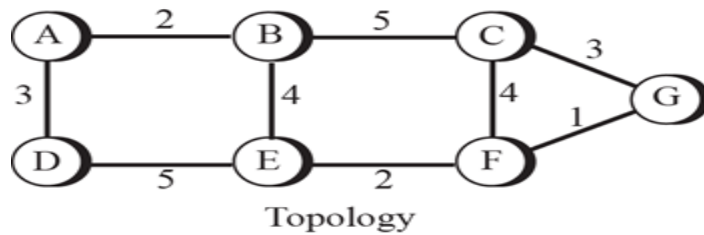
Q1: Apply Dijkstra algorithm to find the set of shortest paths from node 4 to other nodes by considering the network shown below. Show all the steps of Dijkstra algorithm. What is the shortest path tree? **(6+4 =10) [CLO 4]**



Solution: Check all the steps of the algorithm



Q2: Apply the distance-vector algorithm and show the distance table entries at node G (initial one and after 1st iteration) by considering the network shown below. Moreover, assume that each node initially knows the costs to each of its neighbors. (3+2+2+3 = 10) [CLO 4]



Initial Table

Cost to		A	B	C	D	E	F	G
Cost from	C	∞	∞	∞	∞	∞	∞	∞
	F	∞	∞	∞	∞	∞	∞	∞
	G	∞	∞	∞	∞	∞	∞	0

After 1st Iteration

Cost to		A	B	C	D	E	F	G
Cost from	C	∞	5	0	∞	∞	4	3
	F	∞	∞	4	∞	2	0	1
	G	∞	8	3	∞	3	1	0

National University of Computer and Emerging Sciences, Lahore Campus
Quiz5 [BS(CS): Section C] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: December 5, 2022

Total Marks: 20

Duration: 30 -Minutes

Name ----- Roll #----- Section -----

Instructions: Answer all the questions on this sheet. You can make use of rough sheet (not to be attached).

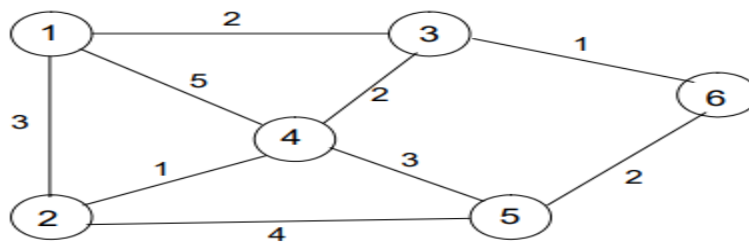
Q 1: A network is using RIP for routing table update. RIP updates routing information periodically. There are N nodes in this network, and each node has B neighbors. **Estimate/discover** the number of message exchanges per second. **(4) [CLO 4]**

Q2: Consider the network shown below:

(a) Apply the Bellman-Ford algorithm to find the set of shortest paths from all nodes to destination node 2.

(b) Apply the same algorithm to find the set of shortest paths from all nodes to destination node 2 after the link between node 2 and 4 goes down.

In both cases, you are simply required to provide the shortest path vector from all nodes to destination node as $X = \{d(1), d(3), d(4), d(5), d(6)\}$ and graph of set of paths to destination 2. **(8+8 =16) [CLO 4]**



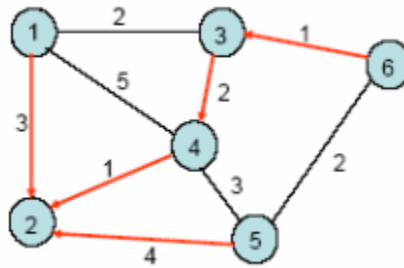
Solution:

Q1: Solution:

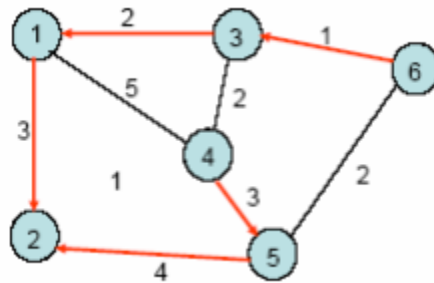
There are N nodes, and each node has B neighbors. RIP updates every 30 seconds, so each node exchanges B messages with its neighbors in 30 seconds. Thus, $NB/30$ messages will be exchanged per second.

Q2:

(a) $X = \{d(1), d(3), d(4), d(5), d(6)\} = \{3, 3, 1, 4, 4\}$ The set of paths to destination 2 are shown below:



(b) $X = \{d(1), d(3), d(4), d(5), d(6)\} = \{3, 5, 7, 4, 6\}$



8.5

National University of Computer and Emerging Sciences, Lahore Campus
Quiz5 [BS(CS): Section D] Fall 2023

Computer Networks (Code: CS3001)

Quiz Date: Novemer 23, 2023

Total Marks: 15

[CLO 3]

Duration: 25 -Minutes

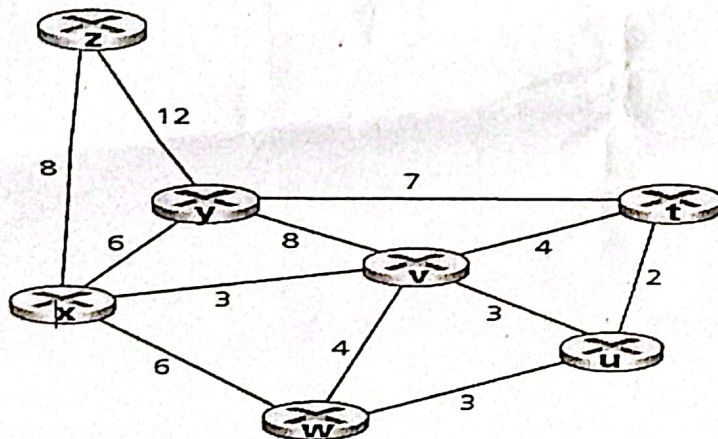
Name ----- Roll # ----- Section -----

Q1: What does BGP stand for? Explain the primary purpose of BGP in networking. [0.5+1.5=2]

Q 2: What is the difference between eBGP and iBGP? [2]

Q3: Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from z to all network nodes. Show the working of the algorithm in the form of a tabular summary for all iterations (i.e., distance and predecessor for all network nodes during each iteration). Moreover, provide the least-cost-path tree from node z and resulting forwarding table in z.

[7+2+2=11]

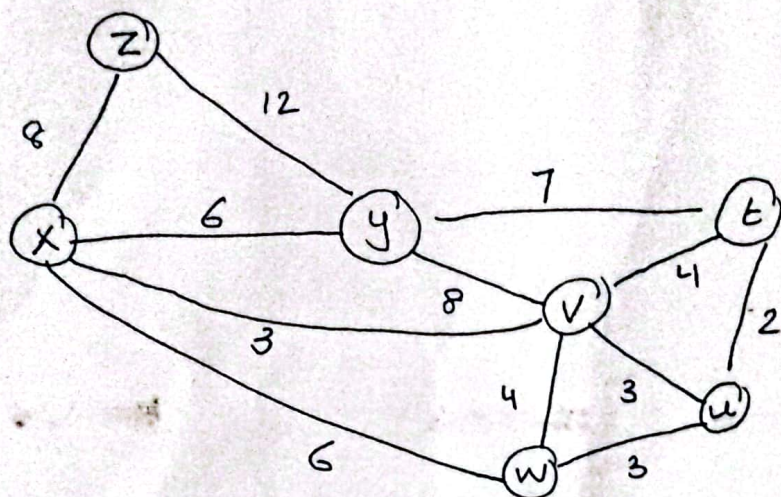


Start writing your answers from here onwards. You can use backside of this sheet if needed.

Q. 1 Ans. BGP stands for Border Gateway Protocol. The primary purpose of BGP in networking is to network is that it is inter-domain routing protocol which allows subnets to broadcast itself and to where it can reach to the Internet.

Q. 2 Ans. eBGP is external BGP where the that it receives network reachability from nearby autonomous systems. iBGP is that it broadcasts the reachability data to all autonomous internal routers.

Q.3
Ans.



1

Step	Starting	D_{zx}	D_{zy}	D_{ze}	D_{zv}	D_{zw}	D_{zu}
Step 0	Z	(8, Z)	12, Z	∞	∞	∞	∞
Step 1	ZX	8, Z	12, Z	∞	(11, X)	14, X	∞
Step 2	ZXV	8, Z	(12, Z)	15, V	11, X	14, X	14, V
Step 3	ZXY	8, Z	12, Z	15, V	11, X	14, X	14, V

✓ 7

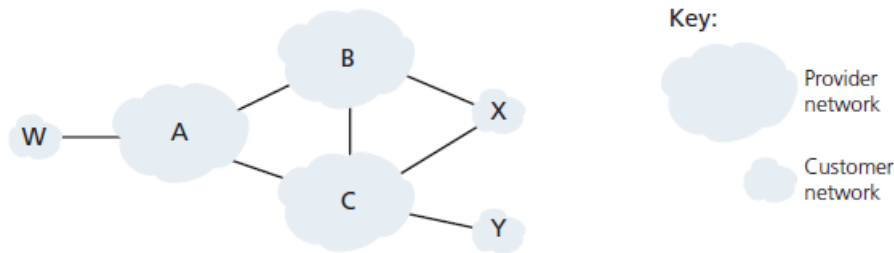
National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	28-November-2023	Section	5J
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

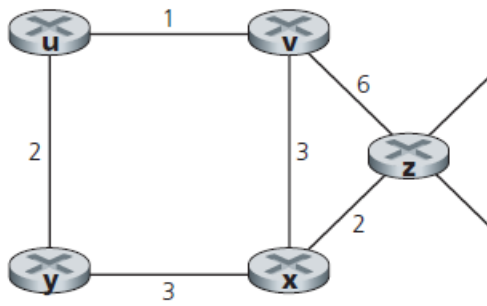
Student Name

Roll No.

Section:



Q1. In Figure above, suppose that there is another stub network V that is a customer of ISP A. Suppose that B and C have a peering relationship, and A is a customer of both B and C. Suppose that A would like to have the traffic destined to W to come from B only, and the traffic destined to V from either B or C. How should A advertise its routes to B and C? What AS routes does C receive? **[5 Marks] [CLO 3]**



Q2. Consider the network shown below, and assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance table entries at node z. **[10 Marks] [CLO 3]**

Solution Quiz#5(5J)

Q1) J

A should advertise to B two routes, AS-paths A-W and A-V.

A should advertise to C only one route, A-V.

C receives AS paths: B-A-W, B-A-V, A-V.

Q2) J

		Cost to				
		u	v	x	y	z
From	v	∞	∞	∞	∞	∞
	x	∞	∞	∞	∞	∞
	z	∞	6	2	∞	0

		Cost to				
		u	v	x	y	z
From	v	1	0	3	∞	6
	x	∞	3	0	3	2
	z	7	5	2	5	0

		Cost to				
		u	v	x	y	z
From	v	1	0	3	3	5
	x	4	3	0	3	2
	z	6	5	2	5	0

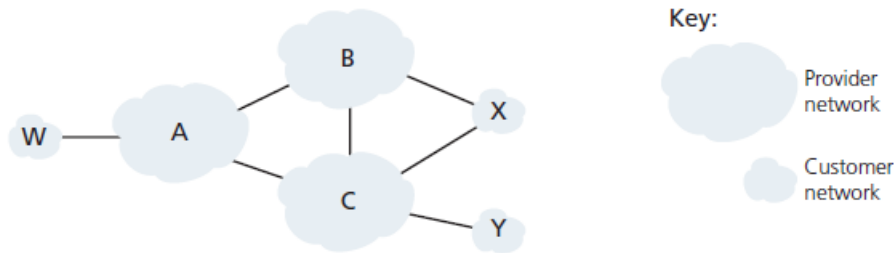
		Cost to				
		u	v	x	y	z
From	v	1	0	3	3	5
	x	4	3	0	3	2
	z	6	5	2	5	0

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	28-November-2023	Section	5K
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

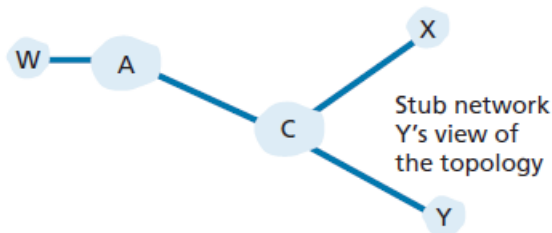
Student Name

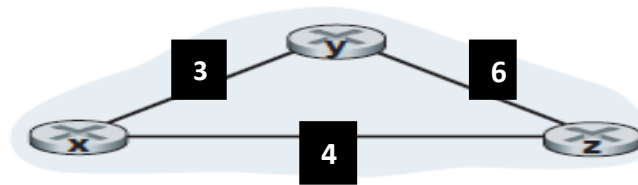
Roll No.

Section:



Q1. In Figure above, consider the path information that reaches stub networks W, X, and Y. Based on the information available at W and X, what are their respective views of the network topology? Justify your answer. The topology view at Y is shown below. **[5 Marks] [CLO 3]**

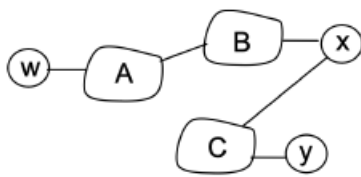




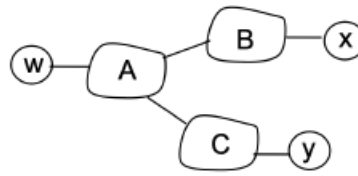
Q2. Compute the distance tables for x, y and z after the initialization step and after each iteration of a synchronous version of the distance-vector algorithm. **[10 Marks] [CLO 3]**

Solution Quiz#5(5K)

Q1)



X's view of the topology



W's view of the topology

In the above solution, X does not know about the AC link since X does not receive an advertised route to w or to y that contain the AC link (i.e., X receives no advertisement containing both AS A and AS C on the path to a destination).

Q2)

Node x table

		Cost to		
		x	y	z
From	x	0	3	4
	y	∞	∞	∞
	z	∞	∞	∞

		Cost to		
		x	y	z
From	x	0	3	4
	y	3	0	6
	z	4	6	0

Node y table

		Cost to		
		x	y	z
From	x	∞	∞	∞
	y	3	0	6
	z	∞	∞	∞

		Cost to		
		x	y	z
From	x	0	3	4
	y	3	0	6
	z	4	6	0

Node z table

		Cost to		
		x	y	z
From	x	∞	∞	∞
	y	∞	∞	∞

z 4 6 0

		Cost to		
		x	y	z
From	x	0	3	4
	y	3	0	6
	z	4	6	0

National University of Computer and Emerging Sciences
Lahore Campus

Computer Networks

(CS3001)

Date: December 06, 2024

Course Instructor

Ms. Saba Tariq

Quiz 5(a)

Total Time (min): 20

Total Marks: 20

Total Questions: 2

Solution

Roll No

Section

Student Signature

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Attempt all the questions.

Q1: You are the network administrator for a small organization that has been assigned the IP address block 192.168.1.0/24. Your task is to subnet this block to fulfill the following requirements:

1. Department A needs 50 hosts.
2. Department B needs 30 hosts.
3. Department C needs 20 hosts.
4. Department D needs 10 hosts.

$$2^5 = 32$$

- Calculate the subnet mask for each department based on the number of hosts required.
- Assign the appropriate subnet range to each department.
- Provide the **network address**, **broadcast address**, and **usable IP range** for each subnet.
- Ensure that no IP address is wasted and that the subnets do not overlap.

A/

A: Network address: 192.168.1.0/26
Usable IP Range: 192.168.1.1 - 192.168.1.62
Broadcast Address: 192.168.1.63/26
Subnet Mask: 255.255.255.192

B: Network add: 192.168.1.64/27
Usable IP Range: 192.168.1.65 - 192.168.1.94
Broadcast add: 192.168.1.95/27
Subnet mask: 255.255.255.224

C: Network add: 192.168.1.96/27
Usable IP Range: 192.168.1.97 - 192.168.1.126
Broadcast add: 192.168.1.127/27
Subnet mask: 255.255.255.224

D: Network add: 192.168.1.128/28
Usable IP Range: 192.168.1.129 - 192.168.1.142
Broadcast add: 192.168.1.143/28
Subnet mask: 255.255.255.240

Computer Networks

(CS3001)

Date: December 06, 2024

Course Instructor

Ms. Saba Tariq

Quiz 5(b)

Total Time (min): 20

Total Marks: 20

Total Questions: 2

Solutions

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions.

Q1: You are the network administrator for a small organization that has been assigned the IP address block 100.16.27.0/25. Your task is to subnet this block to fulfill the following requirements:

1. Department A needs 40 hosts.
2. Department B needs 30 hosts.
3. Department C needs 20 hosts.
4. Department D needs 10 hosts.

- Calculate the subnet mask for each department based on the number of hosts required.
- Assign the appropriate subnet range to each department.
- Provide the network address, broadcast address, and usable IP range for each subnet.
- Ensure that no IP address is wasted and that the subnets do not overlap.

$$\begin{array}{r} 128 \\ 16 \\ \hline 144 \end{array}$$

B/

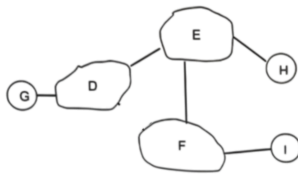
- | | |
|----|---|
| A: | Network add: 100.16.27.0 / 26
IP Range: 100.16.27.1 - 100.16.27.62
Broadcast add: 100.16.27.63 / 26
Subnet mask: 255.255.255.192 |
| B: | Network add: 100.16.27.64 / 27
IP Range: 100.16.27.65 - 100.16.27.94
Broadcast add: 100.16.27.95 / 27
Subnet mask: 255.255.255.224 |
| C: | Network add: 100.16.27.96 / 27
IP Range: 100.16.27.97 - 100.16.27.126
Broadcast add: 100.16.27.127 / 27
Subnet mask: 255.255.255.224 |
| D: | No assignable IP add. for department D. |

SOLUTION QUIZ-05 BCS-5A

QUESTION-01

1. $X \rightarrow AS1 \rightarrow AS4 \rightarrow AS5 \rightarrow AS3 \rightarrow Y$
2. Hot Potato Routing and Inter-AS Routing

QUESTION-02



The D and F connections aren't known to H. So only E-F-I and E-D-G will be advertised to it.

QUESTION-03

1. $A \rightarrow C \rightarrow D$ with total cost 5
- 2.

Forwarding table in A		
Destination	Cost	Next Hop
A	0	—
B	3	B
C	4	C
D	4	B

Forwarding table in C		
Destination	Cost	Next Hop
A	4	A
B	2	D
C	0	—
D	1	D

3.

Forwarding table in A		
Destination	Cost	Next Hop
E	6	B

Forwarding table in B		
Destination	Cost	Next Hop
E	3	D

Forwarding table in C		
Destination	Cost	Next Hop
E	3	D

Forwarding table in D		
Destination	Cost	Next Hop
E	2	E

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	20 minutes	Total Marks:	15
	Paper Date:	13-November-2025	Section	5A , 5C
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

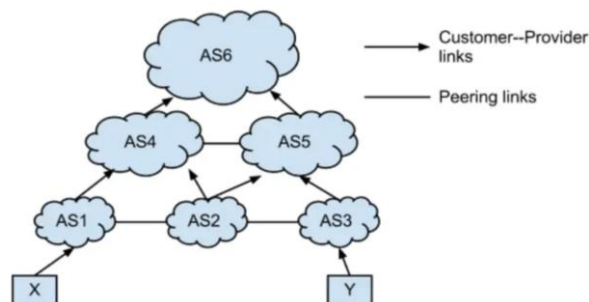
Roll No.

Section:

Q1. Carefully read the Question/Diagram before attempting: [5 marks] [CLO 3]

Consider the AS topology (that uses BGP) shown below. AS1, AS2 and AS3 are access ISP's. AS4 and AS5 are regional ISP's. AS6 is a Tier-1 ISP. Recall the policies AS'es adopt when routing packets (also given in hints below). If there is a tie, assume that AS'es prefer paths with fewer hops. A few helpful hints/rules are as below:

- AS'es prefer customer routes since they are revenue-generating.
- Next in preference are peer routes, which are neutral cost.
- Least preferred are provider routes, as they incur cost.
- BGP prohibits AS'es from forwarding traffic between providers or peers if they are not customers.



1. What path does a packet from host X to host Y take (must be in correct order)?

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2. Considering the path from host X to host Y. Is it an example of **Hot Potato** routing or **Cold Potato** routing or **Neither**? Should this be classified as Intra-AS routing or Inter-AS routing or Neither?

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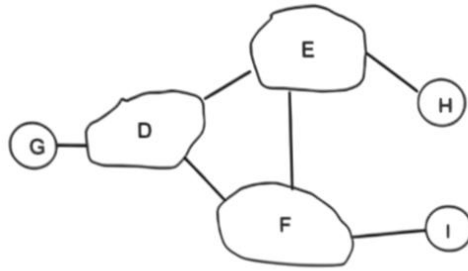
Q2: Draw Diagram and give justification:

[2 Marks] [CLO 1]

In the left Figure, Consider the path information that reaches stub networks G, H and I.

Based on the information available at H. What is its view of Network Topology (for H only)?

Justify your answer. Topological view from G is shown below in right figure:



view from G

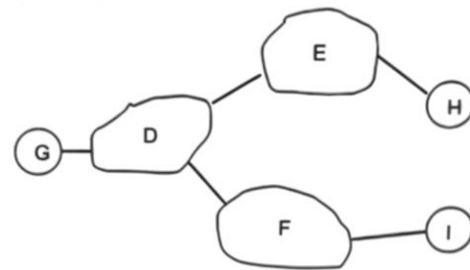


Diagram & Justification:

Q3: Distance-Vector routing Algorithm (Bellman Ford): [1 + 3 + 4 Marks] [CLO 3]

Consider a network where the Distance-Vector algorithm is used. There are 4 routers labeled A, B, C and D. Suppose their routing tables are as follows:

Forwarding table in A		
Destination	Cost	Next Hop
A	0	-
B	3	B
C	4	C
D	5	C

Forwarding table in B		
Destination	Cost	Next Hop
A	3	A
B	0	-
C	4	D
D	3	D

Forwarding table in C		
Destination	Cost	Next Hop
A	4	A
B	4	D
C	0	-
D	1	D

Forwarding table in D		
Destination	Cost	Next Hop
A	5	C
B	3	B
C	1	C
D	0	-

1. If a message originates in A and destination is D. Which path does it take (must be in correct order)? What is the total cost taken from A to D?

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2. If at a point, link cost between B and D changes to 1. Suppose, B and D both notice the link cost change and update their forwarding tables immediately as follows:

Forwarding table in B		
Destination	Cost	Next Hop
A	3	A
B	0	-
C	2	D
D	1	D

Forwarding table in D		
Destination	Cost	Next Hop
A	4	B
B	1	B
C	1	C
D	0	-

Show the forwarding tables of A and C after they receive the distance-vector update message from B and D:

Forwarding table in A		
Destination	Cost	Next Hop
A	0	—
B		
C		
D		

Forwarding table in C		
Destination	Cost	Next Hop
A		
B		
C	0	—
D		

3. Assume that the link cost between B and D stays at 1. If a new router E is added to the network with two links to B and D. The cost between B and E is 5 and the link cost between D and E is 2. **Show the new entries (to destination E) added to forwarding tables A, B, C and D after the distance-vector updates and are stable?**

Forwarding table in A		
Destination	Cost	Next Hop
E		

Forwarding table in B		
Destination	Cost	Next Hop
E		

Forwarding table in C		
Destination	Cost	Next Hop
E		

Forwarding table in D		
Destination	Cost	Next Hop
E		

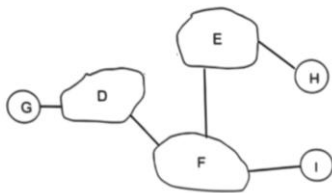
(Note: Feel free to use the rest of the available space/next page for Rough Work)

SOLUTION QUIZ-05 BCS-5C

QUESTION-01

1. AS-F $\rightarrow$ AS-C $\rightarrow$ AS-A $\rightarrow$ AS-D
2. AS-D $\rightarrow$ AS-A $\rightarrow$ AS-B

QUESTION-02



The D - E connection isn't known to F. Thus it can't advertise to I which itself doesn't know. Thus F only advertises F-E-H and F-D-G to I.

QUESTION-03

1.

After first Exchange:

	A	B	C	D	E
A	—	10/B	7/D	3/E	1/E
B	10/A	—	2/C	5/C	11/A
C	7/D	2/B	—	3/D	5/D
D	3/E	5/C	3/C	—	2/E
E	1/A	11/A	5/D	2/D	—

After 2nd Exchange:

	A	B	C	D	E
A	—	8/E	6/E	3/E	1/E
B	8/C	—	2/C	5/C	7/C
C	6/D	2/B	—	3/D	5/D
D	3/E	5/C	3/C	—	2/E
E	1/A	7/D	5/D	2/D	—

After 3rd Exchange:

	A	B	C	D	E
A	—	8/E	6/E	3/E	1/E
B	8/C	—	2/C	5/C	7/C
C	6/D	2/B	—	3/D	5/D
D	3/E	5/C	3/C	—	2/E
E	1/A	7/D	5/D	2/D	—

After 4th Exchange:

	A	B	C	D	E
A	—	8/E	6/E	3/E	1/E
B	8/C	—	2/C	5/C	7/C
C	6/D	2/B	—	3/D	5/D
D	3/E	5/C	3/C	—	2/E
E	1/A	7/D	5/D	2/D	—

(Note: Only the First 2 are required in Question as the Quiz has to fit in 15 minutes)

2.

Forwarding table in B (initial)		
Destination	Cost	Next Hop
A	10	A
C	2	C
D	∞	—
E	∞	—

Forwarding table in B (1 exchange)		
Destination	Cost	Next Hop
A	10	A
C	2	C
D	5	C
E	11	A

Forwarding table in B (2 exchanges)		
Destination	Cost	Next Hop
A	8	C
C	2	C
D	5	C
E	7	C

3.

B → A (directly with cost of 10) instead of the lower cost path

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	20 minutes	Total Marks:	15
	Paper Date:	13-November-2025	Section	5A, 5C
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

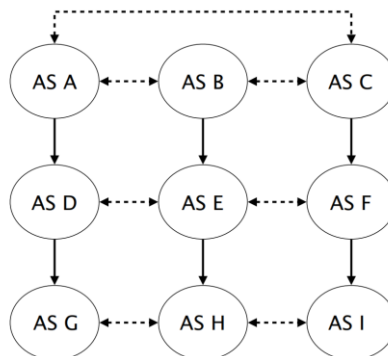
Roll No.

Section:

Q1. Carefully read the Question/Diagram before attempting: [5 marks] [CLO 3]

Consider the AS topology (that uses BGP) shown below. Single-headed plain arrows point from providers to their customers (AS-A is the provider of AS-D), while double-headed dashed arrows connect peers (AS-D and AS-E are peers) Recall the policies AS'es adopt when routing packets (also given in hints below). If there is a tie, assume that AS'es prefer paths with fewer hops. A few helpful hints/rules are as below:

- AS'es prefer customer routes since they are revenue-generating.
- Next in preference are peer routes, which are neutral cost.
- Least preferred are provider routes, as they incur cost.
- BGP prohibits AS'es from forwarding traffic between providers or peers if they are not customers.



1. What path does a packet from AS-F to AS-D take (must be in correct order)?

2. What path does a packet from AS-D to AS-B take (must be in correct order)?

Q2: Draw Diagram and give justification:

[2 Marks] [CLO 3]

In the left Figure, Consider the path information that reaches stub networks G, H and I.

Based on the information available at I. What is its view of Network Topology? Justify your

answer (justification in a few lines). Topological view from G is shown below in right figure:

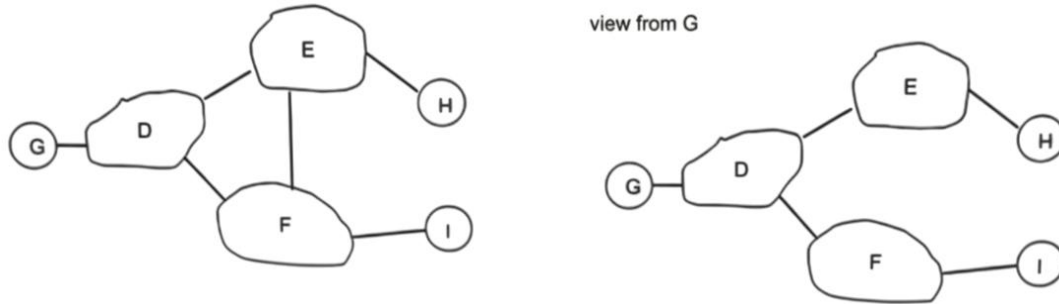
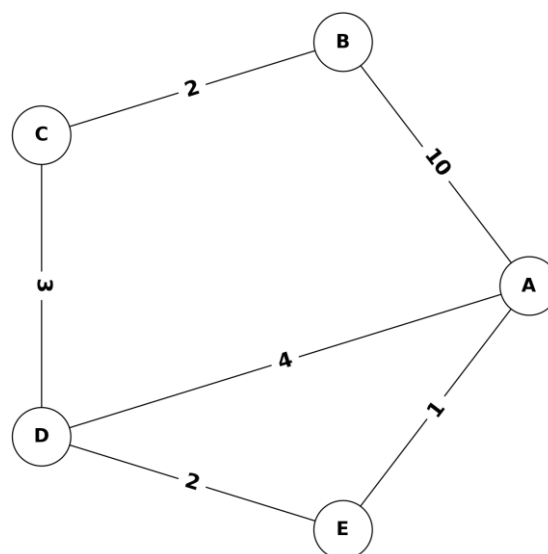


Diagram & Justification:

Q2: Distance-Vector routing Algorithm (Bellman Ford):

[7 + 1 Marks] [CLO 3]



Distance-Vector algorithm has to be performed starting with the initial table shown below, Entries in the table are in the form of **cost/next-hop** (you need to follow the same form). In

this problem node A receives vectors from node B, D and E. Fill in the tables below (the initial table has already been filled for your convenience).

Info at Node	Distance to reach Node (Initial Table)				
	A	B	C	D	E
A	—	10/B	∞	4/D	1/E
B	10/A	—	2/C	∞	∞
C	∞	2/B	—	3/D	∞
D	4/A	∞	3/C	—	2/E
E	1/A	∞	∞	2/D	—

Info at Node	Distance to reach Node (after 1 exchange)				
	A	B	C	D	E
A	—				
B		—			
C			—		
D				—	
E					—

Info at Node	Distance to reach Node (after 2 exchanges)				
	A	B	C	D	E
A	—				
B		—			
C			—		
D				—	
E					—

Forwarding table in B (initial)		
Destination	Cost	Next Hop
A		
C		
D		
E		

Forwarding table in B (1 exchange)		
Destination	Cost	Next Hop
A		
C		
D		
E		

Forwarding table in B (2 exchanges)		
Destination	Cost	Next Hop
A		
C		
D		
E		

Based on the above data, let's say that the algorithm stops after the 1st exchange (before completing a 2nd exchange). What is the path a packet takes when going from B to A (must be the exact path)? What is the total cost?

--	--

(Note: Feel free to use rest of the space/extra space for Rough Work)

SOLUTION QUIZ-05 BCS-6A

QUESTION-01

1. B
2. C
3. C

TRUE/FALSE

1. True
2. False

QUESTION-02

Step	N'	$D(t),p(t)$	$D(u),p(u)$	$D(v),p(v)$	$D(w),p(w)$	$D(y),p(y)$	$D(z),p(z)$
0	x	∞	∞	3,x	6,x	6,x	8,x
1	xv	7,v	6,v	3,x	6,x	6,x	8,x
2	xvu	7,v	6,v	3,x	6,x	6,x	8,x
3	xvuw	7,v	6,v	3,x	6,x	6,x	8,x
4	xvuwy	7,v	6,v	3,x	6,x	6,x	8,x
5	xvuwyt	7,v	6,v	3,x	6,x	6,x	8,x
6	xvuwytz	7,v	6,v	3,x	6,x	6,x	8,x

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	25-Apr-2024	Section	6A
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

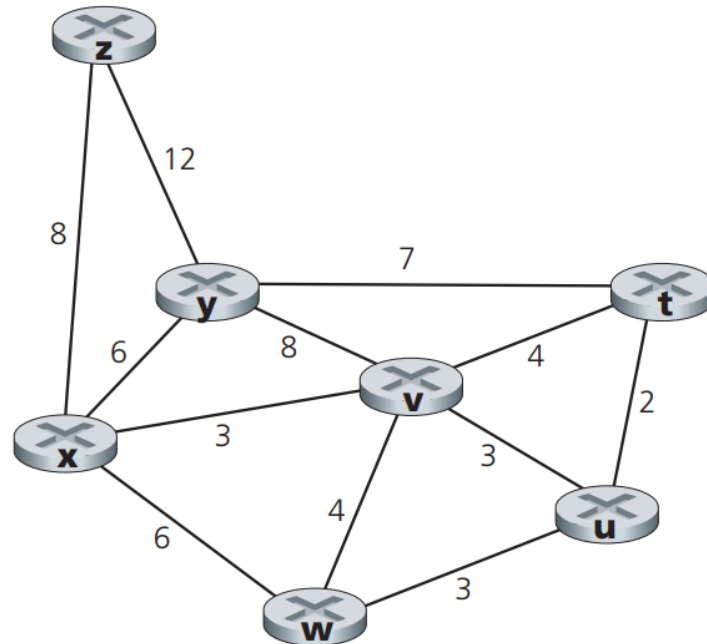
1. What is one use of the AS Path attribute from below?
 - a) To determine if a route is active in the routing table
 - b) To prevent routing loops
 - c) To establish a BGP peer session
 - d) To prevent a denial-of-service attack
2. What is the primary purpose of BGP (Border Gateway Protocol)?
 - a) Exchange routing information within an Autonomous System (AS)
 - b) Facilitate communication between routers within the same subnet
 - c) Connect different autonomous systems and exchange routing information
 - d) Route traffic within a local network
3. What is the key characteristic of a distance vector routing protocol?
 - a) It maintains a complete map of the network.
 - b) It uses a link-state database for routing decisions.
 - c) It periodically shares its entire routing table with its immediate neighbors.
 - d) It relies on policies to determine the best path.

True/False:

4. Hot-potato routing is the practice of passing traffic off to another autonomous system as quickly as possible. [T / F]
5. In a distance-vector routing algorithm, each node has a map of the entire network and determines the shortest path from itself to all other nodes in the network. [T / F]

Question 2: Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from **x** to all network nodes.

[10 Marks] [CLO 3]



Step	N'	D(t),P(t)	D(u),P(u)	D(v),P(v)	D(w),P(w)	D(y),P(y)	D(z),P(z)

SOLUTION QUIZ-05 BSE-6A

QUESTION-01

- 1) B
- 2) A
- 3) C

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

S	B	C	D	E	F	G
A	2,A	6,A	3,A	∞	∞	∞
A, B	-	6,A	3,A	7,B	∞	∞
A, B, D	-	5,D	-	7,B	11,D	∞
A, B, D, C	-	-	-	6,C	11,D	∞
A, B, D, C, E	-	-	-	-	10,E	∞
A, B, D, C, E, F	-	-	-	-	-	11,F

National University of Computer and Emerging Sciences, Lahore Campus

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	Program:	BS (Software Engineering)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	25-Apr-2024	Section	BSE-6A
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. In BGP, what is the purpose of the AS Path attribute?
 - a) Specifies the length of the Autonomous System
 - b) Lists the Autonomous Systems through which the route has passed
 - c) Defines the administrative distance of the route
 - d) Determines the cost of the route

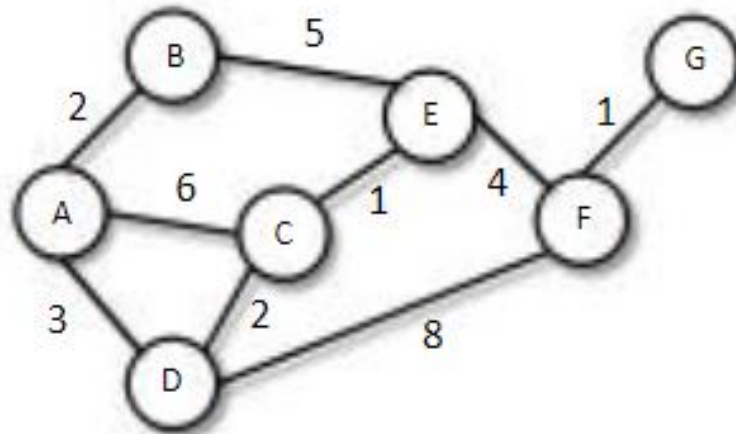
2. In distance vector routing algorithm, the routing tables are updated
 - a) by exchanging information with the neighbours
 - b) automatically
 - c) using the backup database
 - d) by the server

3. How do the routers exchange external routing information between ASs?
 - a) Inject the routes into IGP
 - b) Through IBGP
 - c) Through EBGp
 - d) Advertise local routing information to other AS through static route

True/False:

4. IGP protocols, like Bellman Ford, use Dijkstra's algorithm to calculate the shortest path to a destination. [T / F]
5. It is necessary that every autonomous system use the same intra-AS routing algorithm. [T / F]

Question 2: Consider the network given below. The numbers indicate the link costs. For the indicated link costs, use Dijkstra's algorithm to find the shortest paths from node **A** to all other nodes. **[10 Marks] [CLO 3]**



Step	N'	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)	D(G),P(G)

SOLUTION QUIZ-05 BSE-6C

QUESTION-01

A should advertise to B two routes, AS-paths A-W and A-V.

A should advertise to C only one route, A-V.

C receives AS paths: B-A-W, B-A-V, A-V.

QUESTION-02

		Cost to				
		u	v	x	y	z
From	v	∞	∞	∞	∞	∞
	x	∞	∞	∞	∞	∞
	z	∞	6	2	∞	0

		Cost to				
		u	v	x	y	z
From	v	1	0	3	∞	6
	x	∞	3	0	3	2
	z	7	5	2	5	0

		Cost to				
		u	v	x	y	z
From	v	1	0	3	3	5
	x	4	3	0	3	2
	z	6	5	2	5	0

		Cost to				
		u	v	x	y	z
From	v	1	0	3	3	5
	x	4	3	0	3	2
	z	6	5	2	5	0

**National University of Computer and Emerging Sciences, Lahore
Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	29-April-2025	Section	6A
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

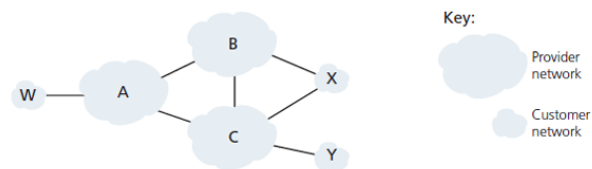
Roll No.

Section:

Q1.

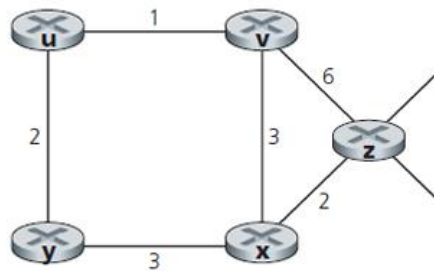
[5 marks] [CLO 3]

In Figure below, suppose that there is another stub network V that is a customer of ISP A. Suppose that B and C have a peering relationship, and A is a customer of both B and C. Suppose that A would like to have the traffic destined to W to come from B only, and the traffic destined to V from either B or C. How should A advertise its routes to B and C? What AS routes does C receive?



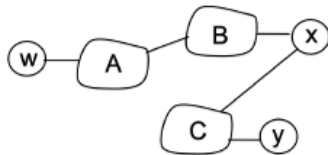
Question 2:**[Marks 10] [CLO 3]**

Consider the network shown below, assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance table entries at node z.

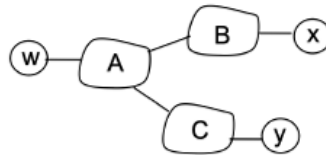


SOLUTION QUIZ-05 BSE-6C

QUESTION-01



X's view of the topology



W's view of the topology

In the above solution, X does not know about the AC link since X does not receive an advertised route to w or to y that contain the AC link (i.e., X receives no advertisement containing both AS A and AS C on the path to a destination).

QUESTION-02

Node x table

		Cost to		
		x	y	z
From	x	0	3	4
	y	∞	∞	∞
	z	∞	∞	∞

		Cost to		
		x	y	z
From	x	0	3	4
	y	3	0	6
	z	4	6	0

Node y table

		Cost to		
		x	y	z
From	x	∞	∞	∞
	y	3	0	6
	z	∞	∞	∞

		Cost to		
		x	y	z
From	x	0	3	4
	y	3	0	6
	z	4	6	0

Node z table

		Cost to		
		x	y	z
From	x	∞	∞	∞
	y	∞	∞	∞
	z	4	6	0

		Cost to		
		x	y	z
From	x	0	3	4
	y	3	0	6
	z	4	6	0

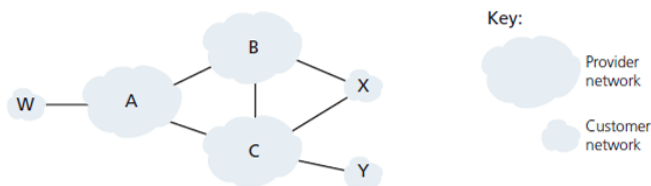
**National University of Computer and Emerging Sciences, Lahore
Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	29-April-2025	Section	6C
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

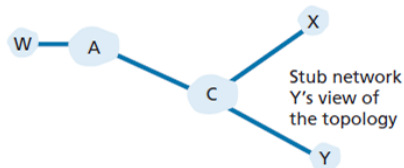
Student Name

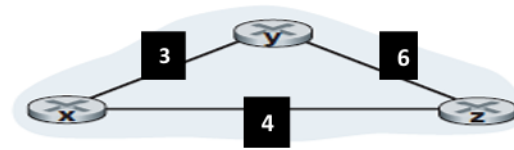
Roll No.

Section:



Q1. In Figure above, consider the path information that reaches stub networks W, X, and Y. Based on the information available at W and X, what are their respective views of the network topology? Justify your answer. The topology view at Y is shown below. **[5 Marks] [CLO 3]**





Q2. Compute the distance tables for x, y and z after the initialization step and after each iteration of a synchronous version of the distance-vector algorithm. **[10 Marks] [CLO 3]**

**National University of Computer and Emerging Sciences,
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	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	23-April-2026	Section	6A , 6B
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

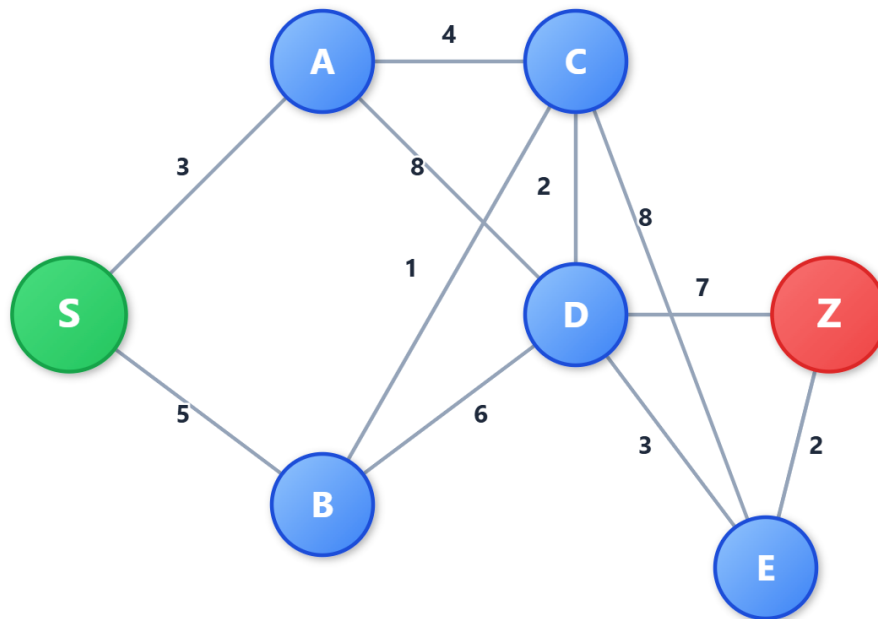
Section:

Q1. Encircle the correct option.

[5 Marks] [CLO 3]

1. BGP is a _____ protocol.
 - a. Distance Vector
 - b. Link State Protocol
 - c. Path Vector Protocol
 - d. None of the above
2. In the context of Link-State routing algorithms, which of the following scenarios is the primary cause of 'routing oscillation'?
 - a. The use of link costs that are dynamically calculated based on traffic volume (load.)
 - b. A topology having multiple equal path costs to the same destination
 - c. A delay in the flooding of Link State Advertisements (LSAs) across the network
 - d. Routing oscillations do not occur in link state algorithms
3. When a router in an Autonomous System (AS) has multiple possible exit points to reach a destination in a different AS, which criterion does it use to implement 'hot potato' routing?
 - a. It randomly alternates between all available paths to all exit points to ensure load balancing
 - b. It selects the path to the exit point with the lowest internal cost (lowest intra-AS cost)
 - c. It selects the shortest path to the final destination
 - d. It opts for cold potato instead of hot potato routing
4. A router learns a route to 10.0.0.0/8 via BGP with Next-Hop 192.0.2.1. However, the router's routing table does not have an entry for 192.0.2.1. What is the result?
 - a. The router installs the route but marks it as "unreachable."
 - b. The router initiates an ARP request for 192.0.2.1.
 - c. The BGP route is considered invalid and is not installed in the forwarding table.
 - d. The router uses the default gateway to forward traffic for this destination.
5. Which property is TRUE about Distance Vector (DV) algorithms?
 - a. Require full topology knowledge
 - b. Use centralized computation
 - c. Converge via iterative neighbor exchange
 - d. Never suffer from routing loops

Q2. Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the link state algorithm and compute shortest path from S to all other nodes by filling the below table step by step. **[10 Marks] [CLO 3]**



Step	Set N'	D(A), p(A)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(Z), p(Z)
0							
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

National University of Computer and Emerging Sciences (Lahore Campus)

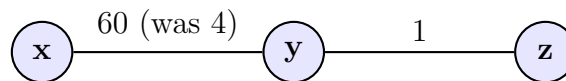
Quiz 5: Network Layer - Control Plane (Chapter 5)

Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) Distance Vector Analysis

Consider the 3-node linear topology below. Bellman-Ford Distance Vector routing is used. Link costs are: $c(x, y) = 4$, $c(y, z) = 1$. At t_0 , the network is stabilized. At t_1 , the link cost $c(x, y)$ dramatically increases to 60. Router y detects this change immediately. Router z does not see the link $x - y$ directly. Assume Poison Reverse is **NOT** enabled.

1. Calculate the Distance Vector entries $D_y(x)$ and $D_z(x)$ for steps t_1, t_2, t_3 .
2. Why does the network not stabilize (What is this problem called)? Give one way it can be prevented?

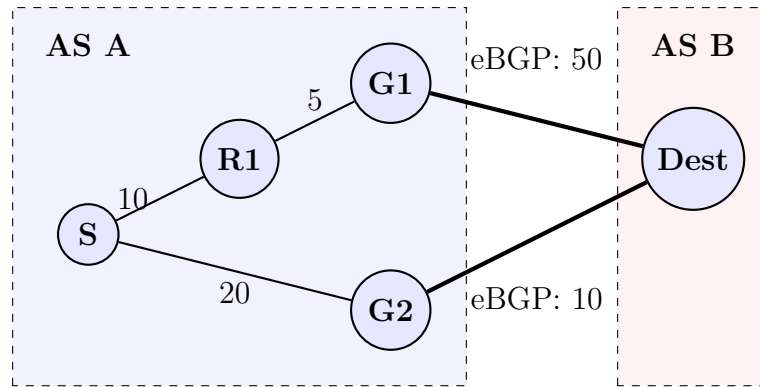


2. (5 points) Hot vs. Cold Potato Routing Calculation

AS A needs to send a packet to destination IP in AS B. AS A has two gateways ($G1, G2$) connected to AS B. The IGP (Intra-AS) costs within AS A are shown in the diagram. The eBGP costs (latency) on the links between AS A and AS B are also shown.

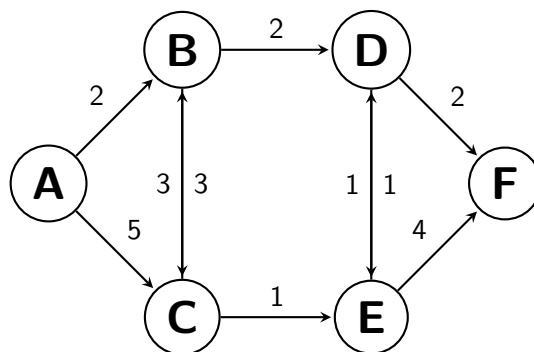
- **Hot Potato Routing:** Minimize intra-AS cost.
- **Cold Potato Routing:** Minimize total cost (Intra + Inter).

Calculate the path and total cost for the source node S for both routing strategies. What is the efficiency penalty (Total Cost Difference) of using Hot Potato routing in this specific topology?



3. (5 points) **Dijkstra's Algorithm and Forwarding tables**

Compute the Forwarding Table for Node **E** (not the source A) using Dijkstra's algorithm. Note the directed edges and asymmetric costs. Run Dijkstra's algorithm originating at Node **E**. Provide the resulting Forwarding Table for Node E. (Format: Destination — Next Hop).



National University of Computer and Emerging Sciences (Lahore Campus)

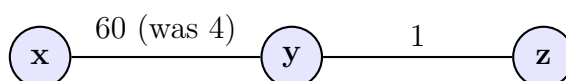
Quiz 5: Network Layer - Control Plane (Chapter 5)

Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) Distance Vector Analysis

Consider the 3-node linear topology below. Bellman-Ford Distance Vector routing is used. Link costs are: $c(x, y) = 4, c(y, z) = 1$. At t_0 , the network is stabilized. At t_1 , the link cost $c(x, y)$ dramatically increases to 60. Router y detects this change immediately. Router z does not see the link $x - y$ directly. Assume Poison Reverse is **NOT** enabled.

1. Calculate the Distance Vector entries $D_y(x)$ and $D_z(x)$ for steps t_1, t_2, t_3 .
2. Why does the network not stabilize (What is this problem called)? Give one way it can be prevented?



Solution:

1. Calculations:

- **Initial (t_0):** $D_y(x) = 4, D_z(x) = 5$ (via y).
- t_1 (**y detects change**): y updates its distance to x.

$$D_y(x) = \min(c(y, x) + D_x(x), c(y, z) + D_z(x)) = \min(60 + 0, 1 + 5) = \mathbf{6}$$

(Note: y chooses path via z because $1 + 5 < 60$. z still thinks dist to x is 5).

- t_2 (**z receives update from y**): z sees $D_y(x) = 6$.

$$D_z(x) = \min(c(z, y) + D_y(x)) = 1 + 6 = \mathbf{7}$$

- t_3 (**y receives update from z**): y sees $D_z(x) = 7$.

$$D_y(x) = \min(60, 1 + 7) = \mathbf{8}$$

2. Problem & Prevention:

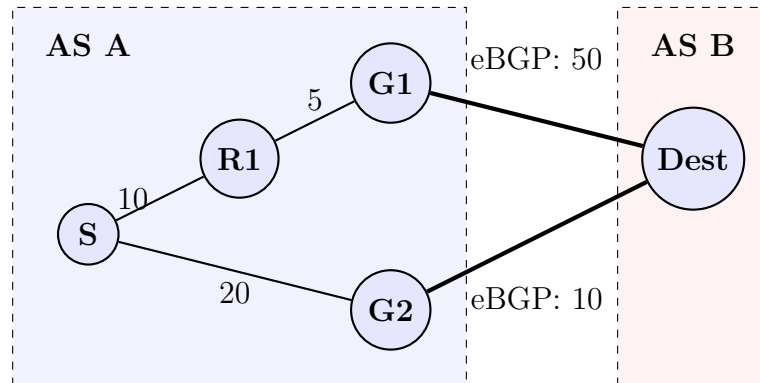
- **Problem: Count-to-Infinity.** The nodes bounce updates back and forth, slowly incrementing the cost until it reaches the direct link cost (60) or infinity.
- **Prevention: Poisoned Reverse.** If z routes to x via y, z should advertise $D_z(x) = \infty$ to y.

2. (5 points) Hot vs. Cold Potato Routing Calculation

AS A needs to send a packet to destination IP in AS B. AS A has two gateways ($G1, G2$) connected to AS B. The IGP (Intra-AS) costs within AS A are shown in the diagram. The eBGP costs (latency) on the links between AS A and AS B are also shown.

- **Hot Potato Routing:** Minimize intra-AS cost.
- **Cold Potato Routing:** Minimize total cost (Intra + Inter).

Calculate the path and total cost for the source node S for both routing strategies. What is the efficiency penalty (Total Cost Difference) of using Hot Potato routing in this specific topology?

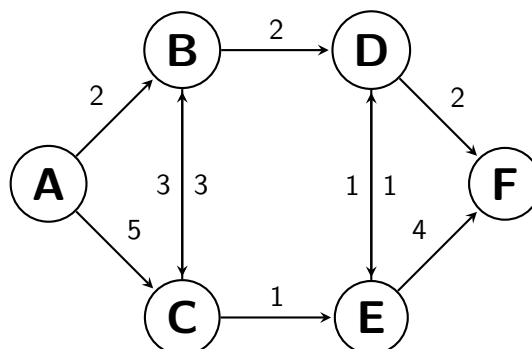


Solution:

- **Internal Costs:** Path to G1 ($S \rightarrow R1 \rightarrow G1$) = $10 + 5 = 15$. Path to G2 ($S \rightarrow G2$) = 20.
- **Hot Potato (Minimize Internal Cost):**
 - Compares internal cost 15 (via G1) vs 20 (via G2). Chooses G1.
 - **Path:** $S \rightarrow R1 \rightarrow G1 \rightarrow \text{Dest}$.
 - **Total Cost:** $15(\text{Intra}) + 50(\text{Inter}) = \mathbf{65}$.
- **Cold Potato (Minimize Total Cost):**
 - Via G1: $15 + 50 = 65$.
 - Via G2: $20 + 10 = 30$.
 - Chooses G2 (since $30 < 65$).
 - **Path:** $S \rightarrow G2 \rightarrow \text{Dest}$.
 - **Total Cost:** **30**.
- **Efficiency Penalty:** $\text{Cost}(\text{Hot}) - \text{Cost}(\text{Cold}) = 65 - 30 = \mathbf{35}$.

3. (5 points) Dijkstra's Algorithm and Forwarding tables

Compute the Forwarding Table for Node **E** (not the source A) using Dijkstra's algorithm. Note the directed edges and asymmetric costs. Run Dijkstra's algorithm originating at Node **E**. Provide the resulting Forwarding Table for Node E. (Format: Destination — Next Hop).



Solution: Running Dijkstra from Source E:

1. **Initialization:** $D(E) = 0$. Neighbors: $D(D) = 1, D(F) = 4$. All others ∞ .
2. **Step 1:** Select node with min dist: **D** (dist=1).
3. **Step 2:** Relax neighbors of D (D goes to E, F).
 - Path to F via D: $D(D) + c(D, F) = 1 + 2 = 3$.
 - Since $3 < 4$ (current $D(F)$), update $D(F)$ to 3. Predecessor of F becomes D.
4. **Step 3:** Select node with min dist: **F** (dist=3). No outgoing edges.
5. Nodes A, B, C are unreachable from E in this directed graph.

Forwarding Table for E:

Destination	Next Hop	Cost
D	D	1
F	D	3
A	- (Unreachable)	∞
B	- (Unreachable)	∞
C	- (Unreachable)	∞

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

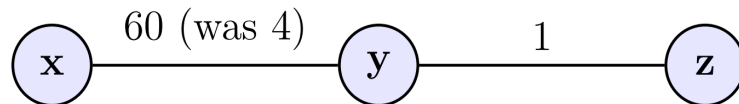
Section:

Q1. Carefully read the Question before attempting:

[5 marks]

Consider the 3-node linear topology below. Bellman-Ford Distance Vector routing is used. Link costs are: $c(x, y) = 4$, $c(y, z) = 1$. At t_0 , the network is stabilized. At t_1 , the link cost $c(x, y)$ dramatically increases to 60. Router y detects this change immediately. Router z does not see the link $x - y$ directly. Assume Poison Reverse is **NOT** enabled.

1. Calculate the Distance Vector entries $D_y(x)$ and $D_z(x)$ for steps t_1 , t_2 , t_3 .
2. Why does the network not stabilize (What is this problem called)? Give one way it can be prevented?



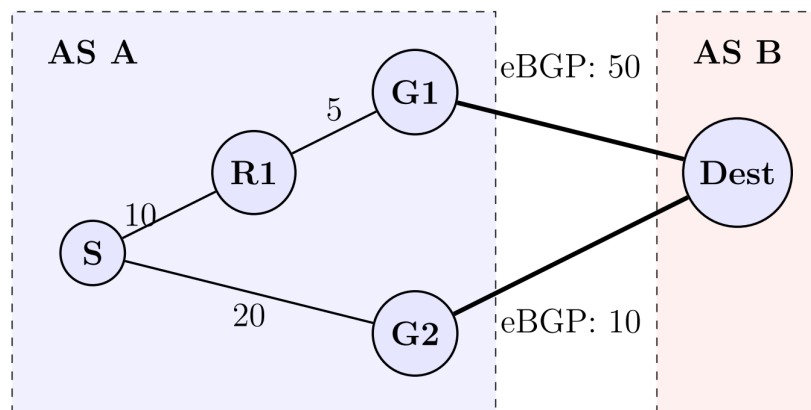
Q2. Carefully read the Question before attempting:

[5 marks]

AS A needs to send a packet to the destination IP in AS B. AS A has two gateways (G_1 , G_2) connected to AS B. The IGP (Intra-AS) costs within AS A are shown in the diagram. The eBGP costs (latency) on the links between AS A and AS B are also shown.

- **Hot Potato Routing:** Minimize intra-AS cost.
- **Cold Potato Routing:** Minimize total cost (Intra + Inter).

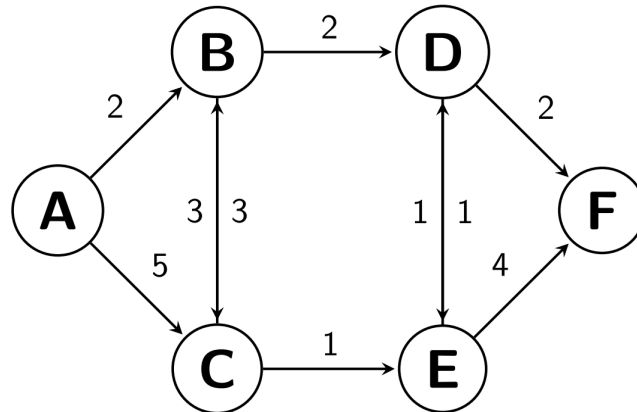
Calculate the path and total cost for the source node S for both routing strategies. What is the efficiency penalty (Total Cost Difference) of using Hot Potato routing in this specific topology?



Q3. Carefully read the Question before attempting:

[5 marks]

Compute the Forwarding Table for **Node E** (not the source A) using Dijkstra's algorithm. Note the directed edges and asymmetric costs. Run Dijkstra's algorithm originating at **Node E**. Provide the resulting Forwarding Table for Node E. (Format: Destination — Next Hop).



National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

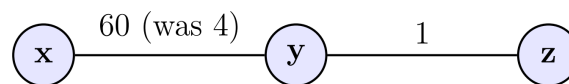
Section:

Q1. Carefully read the Question before attempting:

[5 marks]

Consider the 3-node linear topology below. Bellman-Ford Distance Vector routing is used. Link costs are: $c(x, y) = 4$, $c(y, z) = 1$. At t_0 , the network is stabilized. At t_1 , the link cost $c(x, y)$ dramatically increases to 60. Router y detects this change immediately. Router z does not see the link $x - y$ directly. Assume Poison Reverse is **NOT** enabled.

1. Calculate the Distance Vector entries $D_y(x)$ and $D_z(x)$ for steps t_1 , t_2 , t_3 .
2. Why does the network not stabilize (What is this problem called)? Give one way it can be prevented?



Solution:

1. Calculations:

- **Initial (t_0):** $D_y(x) = 4$, $D_z(x) = 5$ (via y).
- **t_1 (y detects change):** y updates its distance to x.

$$D_y(x) = \min(c(y, x) + D_x(x), c(y, z) + D_z(x)) = \min(60 + 0, 1 + 5) = 6$$

(Note: y chooses path via z because $1 + 5 < 60$. z still thinks dist to x is 5).

- **t_2 (z receives update from y):** z sees $D_y(x) = 6$.

$$D_z(x) = \min(c(z, y) + D_y(x)) = 1 + 6 = 7$$

- **t_3 (y receives update from z):** y sees $D_z(x) = 7$.

$$D_y(x) = \min(60, 1 + 7) = 8$$

2. Problem & Prevention:

- **Problem: Count-to-Infinity.** The nodes bounce updates back and forth, slowly incrementing the cost until it reaches the direct link cost (60) or infinity.
- **Prevention: Poisoned Reverse.** If z routes to x via y, z should advertise $D_z(x) = \infty$ to y.

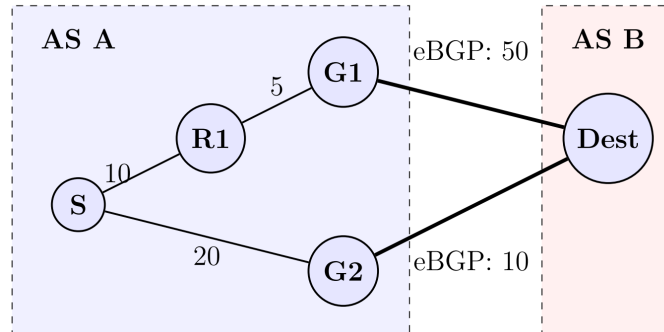
Q2. Carefully read the Question before attempting:

[5 marks]

AS A needs to send a packet to the destination IP in AS B. AS A has two gateways (G1, G2) connected to AS B. The IGP (Intra-AS) costs within AS A are shown in the diagram. The eBGP costs (latency) on the links between AS A and AS B are also shown.

- **Hot Potato Routing:** Minimize intra-AS cost.
- **Cold Potato Routing:** Minimize total cost (Intra + Inter).

Calculate the path and total cost for the source node S for both routing strategies. What is the efficiency penalty (Total Cost Difference) of using Hot Potato routing in this specific topology?



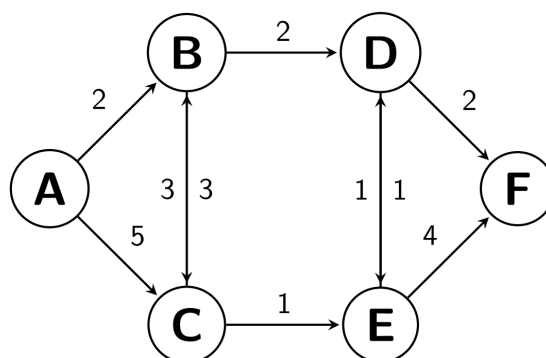
Solution:

- **Internal Costs:** Path to G1 ($S \rightarrow R1 \rightarrow G1$) = $10 + 5 = 15$. Path to G2 ($S \rightarrow G2$) = 20.
- **Hot Potato (Minimize Internal Cost):**
 - Compares internal cost 15 (via G1) vs 20 (via G2). Chooses G1.
 - **Path:** $S \rightarrow R1 \rightarrow G1 \rightarrow \text{Dest}$.
 - **Total Cost:** $15(\text{Intra}) + 50(\text{Inter}) = 65$.
- **Cold Potato (Minimize Total Cost):**
 - Via G1: $15 + 50 = 65$.
 - Via G2: $20 + 10 = 30$.
 - Chooses G2 (since $30 < 65$).
 - **Path:** $S \rightarrow G2 \rightarrow \text{Dest}$.
 - **Total Cost:** 30.
- **Efficiency Penalty:** $\text{Cost}(\text{Hot}) - \text{Cost}(\text{Cold}) = 65 - 30 = 35$.

Q3. Carefully read the Question before attempting:

[5 marks]

Compute the Forwarding Table for **Node E** (not the source A) using Dijkstra's algorithm. Note the directed edges and asymmetric costs. Run Dijkstra's algorithm originating at **Node E**. Provide the resulting Forwarding Table for Node E. (Format: Destination — Next Hop).



Solution: Running Dijkstra from Source E:

1. **Initialization:** $D(E) = 0$. Neighbors: $D(D) = 1, D(F) = 4$. All others ∞ .
2. **Step 1:** Select node with min dist: **D** (dist=1).
3. **Step 2:** Relax neighbors of D (D goes to E, F).
 - Path to F via D: $D(D) + c(D, F) = 1 + 2 = 3$.
 - Since $3 < 4$ (current $D(F)$), update $D(F)$ to 3. Predecessor of F becomes D.
4. **Step 3:** Select node with min dist: **F** (dist=3). No outgoing edges.
5. Nodes A, B, C are unreachable from E in this directed graph.

Forwarding Table for E:

Destination	Next Hop	Cost
D	D	1
F	D	3
A	- (Unreachable)	∞
B	- (Unreachable)	∞
C	- (Unreachable)	∞

SOLUTION QUIZ-05 BSE-6A

QUESTION-01

1. C
2. C
3. B
4. C
5. C

QUESTION-02

Step	Set N'	D(A), p(A)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(Z), p(Z)
0	{S}	4, S	2, s	∞	∞	∞	∞
1	{S,B}	4, S		∞	10, B	12, B	∞
2	{S,B,A}			6, A	9, A	12, B	∞
3	{S,B,A,C}				7, C	12, B	14, C
4	{S,B,A,C,D}					9, D	13, D
5	{S,B,A,C,D,E}						12, E
6	{S,B,A,C,D,E,Z}						

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	23-April-2026	Section	6A , 6B
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

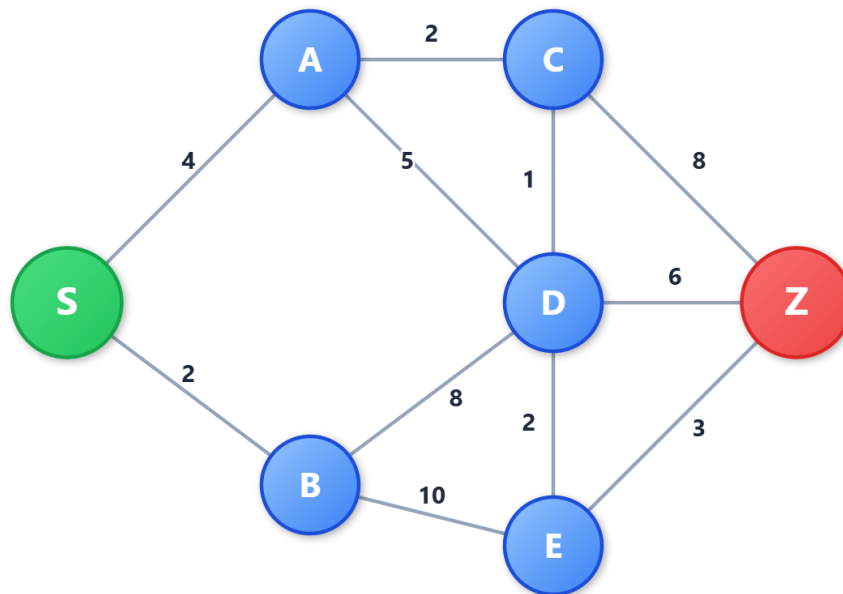
Section:

Q1. Encircle the correct option.

[5 Marks] [CLO 3]

1. In a network of three nodes (A, B, and C), suppose the shortest path from Node B to Node A currently goes through Node C. If the network is implementing the Distance Vector algorithm with Poison Reverse, what value will Node B advertise to Node C as its cost to Node A?
 - a. The actual cost from B to A.
 - b. Cost from B to C plus cost from C to A
 - c. Infinity (∞)
 - d. Zero (0)
2. What is the stopping criteria of a trace route?
 - a. The last router instructs the source to stop the trace route
 - b. Type 11 Code 0 error message is received
 - c. Type 3 Code 3 error message is received
 - d. TTL expires
3. Dominant factor in inter vs intra domain routing respectively is:
 - a. Performance vs Policy
 - b. Policy vs Performance
 - c. Scale vs Policy
 - d. Policy vs Scale
4. Which scenario best represents a link-state algorithm assumption?
 - a. Nodes only know neighbors initially
 - b. Nodes iteratively exchange distance vectors
 - c. Each node has complete knowledge of network topology
 - d. Routing decisions depend on policy only
5. What is the main role of BGP AS-PATH attribute?
 - a. Measures bandwidth
 - b. Stores sequence of routers
 - c. Prevents loops and indicates traversed path
 - d. Determines packet priority

Q2. Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the link state algorithm and compute shortest path from S to all the other nodes by filling the table below step by step. **[10 Marks] [CLO 3]**



Step	Set N'	D(A), p(A)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(Z), p(Z)
0							
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

SOLUTION QUIZ-05 BSE-6B

QUESTION-01

1. C
2. A
3. B
4. C
5. C

QUESTION-02

Step	Set N'	D(A), p(A)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(Z), p(Z)
0	{S}	3, S	5, S	∞	∞	∞	∞
1	{S,A}		5,S	7,A	11,A	∞	∞
2	{S,A,B}			6, B	11,A	∞	∞
3	{S,A,B,C}				8, C	14,C	∞
4	{S,A,B,C,D}					11, D	15, D
5	{S,A,B,C,D,E}						13, E
6	{S,A,B,C,D,E,Z}						